

Project Design Phase-II

Technology Stack

(Architecture & Stack)

Date	27 September 2026
Team ID	[SWTID-2026-9584]
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum Marks	4 Marks

Technical Architecture:

The diagram below shows the layered architecture for the standard laptop procurement automation, followed by the component and characteristic details.

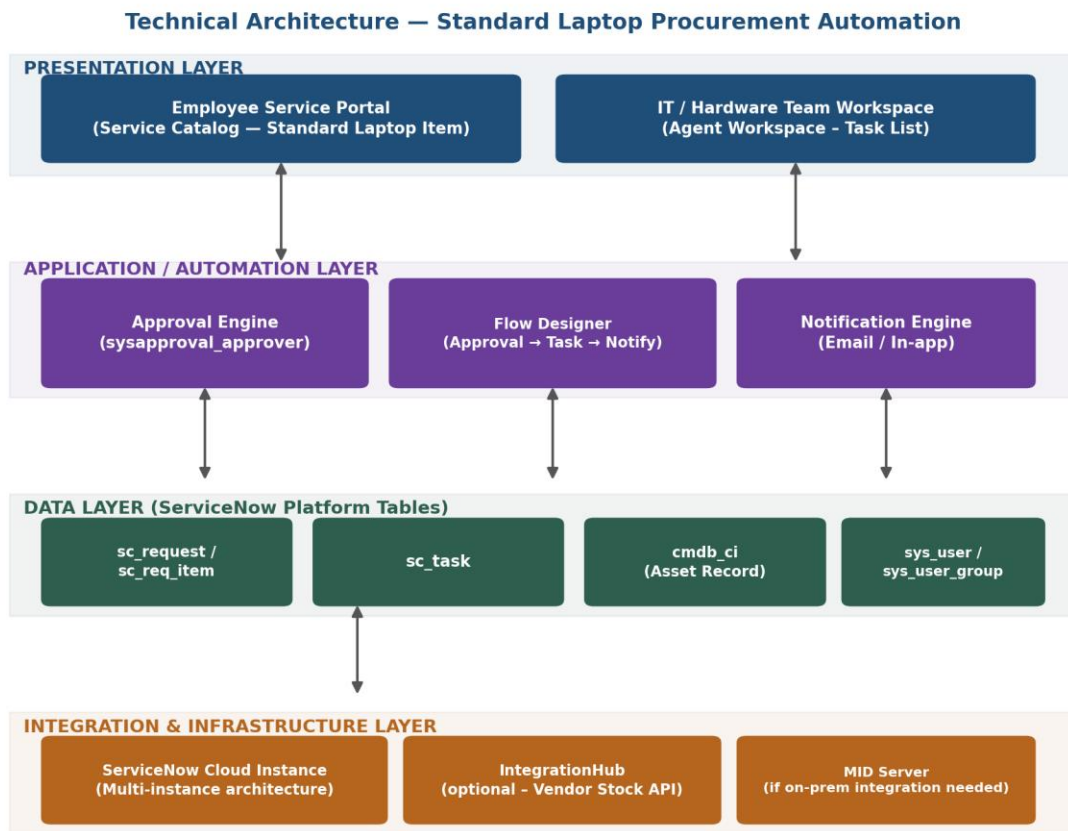


Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	How users interact with the application — request & track laptop orders	ServiceNow Service Portal / Employee Center, Now Experience UI

2	Application Logic-1	Handles catalog form logic, validations & auto-fill	Catalog Client Scripts, UI Policies (GlideScript)
3	Application Logic-2	Manages manager approval routing & decisions	ServiceNow Approval Engine (sysapproval_approver)
4	Application Logic-3	Orchestrates the automation: trigger → approval → task → notify	Flow Designer (Flows, Actions, Subflows)
5	Database	Stores requests, tasks and configuration data	ServiceNow platform tables (sc_request, sc_req_item, sc_task)
6	Cloud Database	Underlying data store for the ServiceNow instance	ServiceNow multi-instance cloud database
7	File Storage	Stores attachments such as configuration checklists	ServiceNow Attachments (sys_attachment)
8	External API-1	(Phase-2) Real-time stock / delivery ETA from vendor	Vendor REST API via IntegrationHub
9	External API-2	Sends email / chat notifications to users	ServiceNow Notification Engine, MS Teams / Slack (optional)
10	Machine Learning Model	Not used in Phase-1; future scope: auto-categorize requests	ServiceNow Predictive Intelligence (future)
11	Infrastructure (Server / Cloud)	Build, test & run the automation	ServiceNow PDI (dev/test), Production cloud instance

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	ServiceNow is a proprietary low-code platform; the catalog UI renders via the Now Experience Framework	Now Experience UI Framework (Angular-based, internal)
2	Security Implementations	Access restricted by role; SSO login; every flow action is logged	ACLs, SSO / OAuth, sys_audit logging
3	Scalable Architecture	Built on ServiceNow's horizontally scalable multi-instance cloud; automation split into reusable subflows	ServiceNow Cloud Architecture, Flow Designer Subflows
4	Availability	Inherits ServiceNow's platform uptime SLA; the flow retries automatically on transient failure	ServiceNow 99.8% Uptime SLA, Flow Engine auto-retry
5	Performance	Flow actions run asynchronously so they don't block the requester; catalog scripts avoid heavy synchronous queries	Async Flow Engine, optimized GlideRecord queries

References

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